

STEPCLIMBER · DESIGN DECISION RECORD

StepClimber TRL 2 review decisions

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DOCUMENT CONTROL

Document ID	SCM-DDR-001	Version	0.2
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/stepclimber		
Source	docs/decisions/0001-trl2-review-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	Record Amish's 2026-09-25 decisions on the TRL 2 review, the items still open, and the new TRL 3 proposals

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted (items 1 to 8, and items 11 to 13 through SCM-DDR-002); items 9 and 10 remain proposed, awaiting Amish

■ Context

The TRL 2 review note (docs/REVIEW.md , session "/populate to a strong TRL 2") and the key design choices in SCM-PRC-001 v0.2 listed the StepClimber design choices as "Proposed, awaiting Amish", most with a recommendation. On 2026-09-25 Amish reviewed the review points for every portfolio repo and wrote: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them." He also approved three cross-cutting SwapCell interface additions (a wake method for hosts without CAN, a charge-while-discharging mode and a latch vibration rating for vehicles), a rule that shared SwapCell packs are priced once and excluded from each dependent kit budget, and a rule that co-design partners are chosen per area later.

This record lists what that instruction decides and what it leaves open because there was no recommendation to accept. TRL 4 is on hold by Amish's instruction.

■ Options considered

The options for items 1 to 8 are in docs/REVIEW.md (TRL 2 session) and SCM-PRC-001 v0.2 and are not repeated here. The options for the new items 11 to 13 are in SCM-CAL-001 v0.1.

■ Decision

Table 1. Decided items. Each is "Decided by Amish, 2026-09-25: go with recommendation".

#	Item	Decision	Where it now lives
1	Meaning of the pitch's tilt sensor	Option A: operator-in-the-loop tilt control (speed shaping, tilt window, stop and alert); no actuated load platform. Pitch reworded to "a tilt sensor that helps the courier hold the load angle steady"	<code>project.yaml</code> , <code>README.md</code> , SCM-PRC-001 v0.3
2	Handle height (R11)	Option A: fixed handle for the first prototype, R11 relaxed to an upright height of 1,500 mm or less. Option C (folding hinge) studied at TRL 3 and not adopted	SCM-REQ-001 v0.3 R11; SCM-CAL-001 section 9
3	Rated stair load	60 kg (132 lb) payload until couriers are asked	SCM-REQ-001 v0.3 R1
4	Climbing mechanism	Tri-star clusters rather than a stepping arm or tracks	SCM-PRC-001 v0.3
5	Holding	Self-locking worm gearmotor plus a spring-applied brake (two independent holds) rather than an efficient gearbox with a brake alone	SCM-PRC-001 v0.3; SCM-CAL-001 section 6
6	Battery	24 V LiFePO4, 8S 10 Ah, rather than the 48 V SwapCell pack	SCM-PRC-001 v0.3; R13
7	Operator position	The courier always stands uphill of the truck, going up and down; stated on labels and in the user guide	SCM-PRC-001 v0.3 safety
8	Budget	No change: <code>budget_usd</code> stays at \$600; the priced BOM is \$580	<code>project.yaml</code> ; SCM-REQ-001 R12

Consequences of the cross-cutting decisions for this repo:

- **SwapCell interface v0.3 items** (wake method, charge-while-discharging, latch vibration rating) do not apply, because item 6 keeps StepClimber on its own 24 V pack.
- **Shared SwapCell pack pricing** does not apply for the same reason; the pack stays in this BOM.
- **Co-design partners** are chosen per area later (item 10 stays open).

Table 2. Items open when this record was written. Items 11 to 13 were decided later on 2026-09-25 (SCM-DDR-002).

#	Item	Status
9	First user group for interviews: gig couriers, a parcel company pilot, or appliance and moving crews	Proposed, awaiting Amish (no recommendation was made)
10	Partner courier group or co-design partner	Proposed, awaiting Amish (to be chosen per area later)
11	Cluster and drive rework to meet R2 and R15 (SCM-CAL-001 section 3). Options: (A) 150 mm arms, 200 mm wheels, a two-stage chain drive with an 08B 20-tooth final sprocket, and a 54 mm shaft-line envelope; this likely pushes R5 past 25 kg and R12 past \$600 and needs a larger motor or 18.8 steps/min. (B) Keep the cluster, restrict R2 to square nosings and find a shaft-line drive within a 48 mm radius (none identified at 146 N·m). (C) Change to a stepping-arm or tracked mechanism, which reverses item 4. Recommendation: A, with R5, R12 and R3 revisited once the parts are priced	Decided by Amish, 2026-09-25: go with recommendation (SCM-DDR-002)
12	Tilt window (R6, R7). Options: tighten the stop window from plus or minus 8 to plus or minus 6 degrees so the grip force stays under 100 N, or relax R6 to about 110 N. Recommendation: tighten to plus or minus 6 degrees	Decided by Amish, 2026-09-25: go with recommendation (SCM-DDR-002)
13	Follow-on target changes if item 11 A is chosen: R5 mass, R12 budget and R3 speed or motor size	Decided by Amish, 2026-09-25: go with recommendation (SCM-DDR-002): R3 17 steps/min, R5 27 kg, R12 \$650

■ Consequences

- SCM-PRC-001, SCM-REQ-001 and SCM-PRB-001 move to v0.3 with these decisions; the design choices in items 1 to 8 are no longer "proposed".
- At v0.1 the parametric model, drawing SCM-DWG-001 Rev P1 and BOM stayed at the baseline cluster and drive. After SCM-DDR-002 they carry the decided rework (drawing Rev P2).
- TRL 3 is the hard stop. Nothing in this record starts TRL 4 work.